

FortiAP Configuration through a FortiGate-40f and Applying WPA2 Personal and Enterprise Security



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05/08/2026

Purpose:

The purpose of this lab is to set up a FortiAP with minimal configuration of the FortiAP, relying almost completely on the FortiGate 40f's capability to act as a wireless LAN Controller to control FortiAPs to configure simple commands like setting security protocols, creating SSIDs, and to set or

change channels for different radios on the AP without having to access the AP.

Background Information:

Wireless networking in enterprise and school environments has begun to require centralized management, segmented network access, and advanced authentication mechanisms to meet the demands of modern security standards. Currently, most of the enterprise wireless infrastructure built based on the CAPWAP protocol (Control and Provisioning of Wireless Access Points), which defines how wireless access points communicate with a central wireless controller. Rather than operating independently, some APs that are capable of utilizing a controller-based network such as the FortiAP-421E can offload its configuration and management entirely to a controller (in this case the FortiGate 40F firewall), which pushes radio settings, SSID configurations, and security policies down to the AP over an encrypted channel. This centralized model allows administrators to manage multiple access points from a single interface, applying consistent policies across the entire wireless infrastructure. This makes it simple for administrators to control, manage, and fix issues when it occurs.

Network segmentation through VLANs plays a critical role in enterprise wireless deployments. By assigning different VLANs to different SSIDs, administrators can separate traffic from different users even if they share the same physical infrastructure. In this lab, VLAN 10 is dedicated to AP management traffic, ensuring that CAPWAP control communications between the FortiAP and the FortiGate remain isolated from client data traffic, while VLAN 20 carries wireless client traffic. The switch acts as the intermediary, tagging and routing traffic between the AP and the firewall according to these VLAN assignments. Without proper VLAN configuration on both the switch and the firewall's subinterfaces, the AP would be unable to reach the wireless controller even if physical connectivity is present.

Authentication is another foundational yet highly important layer of enterprise wireless security. One protocol, WPA2-Personal, also known as WPA2-PSK, relies on a pre-shared passphrase or key that clients would use to connect to the network with. While simple to deploy, this model offers limited accountability and traceability since all users share the same credentials. A different authentication method, WPA2-Enterprise, addresses this limitation by integrating and incorporating with a RADIUS (Remote Authentication Dial-In User Service) server, which is a centralized authentication service defined

by the Remote Authentication Dial-In User Service protocol. Under WPA2-Enterprise, each user authenticates with their own unique login and password credentials through the 802.1X authentication framework, where the access point acts as a forwarding media that transfers credential requests from clients to the RADIUS server, which then grants or denies access based on configured users and passwords on the server. This provides the ability of per-user authentication, making it far more suitable for environments where accountability and access control are required.

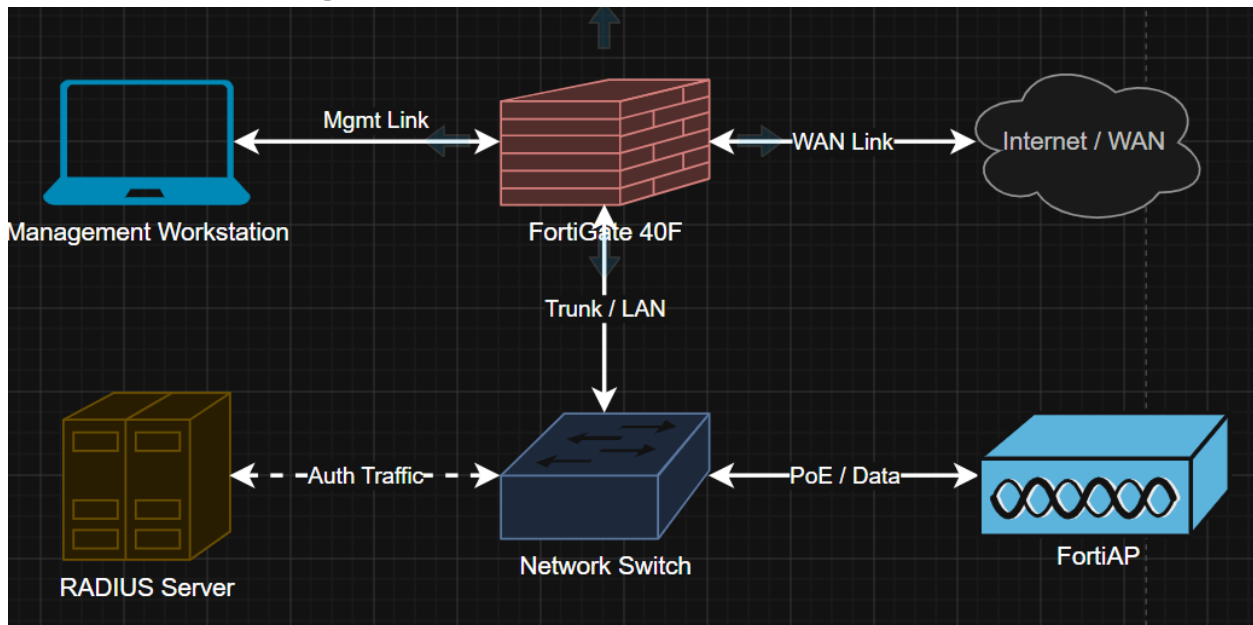
FreeRADIUS is one of the most widely deployed open-source implementations of the RADIUS protocol and serves as the authentication backend in this lab. Running on a Linux machine, FreeRADIUS receives authentication requests forwarded by the FortiGate, validates them against a local user file, and returns either an Access-Accept or Access-Reject response. The shared secret configured between the FortiGate and FreeRADIUS ensures that only trusted network devices can submit authentication requests to the server, preventing unauthorized systems from accessing the User database.

Lab Summary:

In this lab, we deployed a complete enterprise wireless network using a FortiAP-421E access point managed by a FortiGate 40F firewall, with two SSIDs offering different security models and a FreeRADIUS server providing 802.1X authentication. We began by installing and configuring FreeRADIUS on a Linux machine, defining the FortiGate as a trusted RADIUS client in `clients.conf` with a shared secret, and adding test user credentials to the `/users` file. On the Extreme X450E switch, we created and tagged two VLANs. We set VLAN 10 for AP management and VLAN 20 for wireless clients. Then we configured the uplink port to the FortiGate as a trunk carrying both VLANs while the AP-facing port was set as an untagged access port on VLAN 10. On the FortiAP, we configured static controller discovery by setting `AC_DISCOVERY_TYPE` to 1 and pointing `AC_IPADDR_1` to the FortiGate's LAN IP, cleared the DNS hostname to prevent discovery loops, and then saved the configuration. On the FortiGate, we created two VLAN subinterfaces on port A (port facing the switch) corresponding to the switch VLANs, enabled Security Fabric Connection on both the A interface itself and the AP Management interface to allow the wireless controller to bind to it and accept CAPWAP connections, configured a RADIUS server object pointing to the Linux machine, and created two SSIDs. One of which utilizes WPA2-Personal with a pre-shared password and one using WPA2-Enterprise backed by the RADIUS

server with intra-SSID traffic blocking enabled on both. We then built a FortiAP profile assigning both SSIDs to both radios and authorizing the AP under Managed FortiAPs, after which the AP came online and began broadcasting both networks to clients in range.

Network Diagram:



Steps and Configuration:

Step 1: Setting up a FreeRadius Radius Server on a Linux Machine:

1. Run these commands to install FreeRadius onto your Linux device:

```
sudo apt update
```

```
sudo apt install freeradius freeradius-utils
```

2. Open clients.conf with nano and add your FortiGate Firewall as a client at the very bottom of the clients.conf file/

```
sudo nano /etc/freeradius/3.0/clients.conf
```

3. Add this at the very bottom of the clients.conf file

```
client fortigate {  
    ipaddr = LAN-IP-ADDRESS  
    secret = password
```

```
    shortname = fortigate
}
```

4. Open the users file with nano:

```
sudo nano /etc/freeradius/3.0/users
```

5. Add users at the very top (password is CASE SENSITIVE and requires Tab between the username and "Cleartext-Password"):

```
testuser    Cleartext-Password := "Password"
student     Cleartext-Password := "Cisco123"
```

6. Then enable and start FreeRadius:

```
sudo systemctl enable FreeRadius
sudo systemctl start FreeRadius
```

7. Then test the FreeRadius Server with:

```
radtest testuser Password 127.0.0.1 0 testing123
```

Step 2: Configure the Switch to set up different VLANs to allow for AP and various local LAN user connections. Note: for this lab, the switch utilized is an Extreme X450E switch, which has different cli commands than Cisco switches.

1. create the VLANs with the create vlan command:

```
create vlan AP
```

```
create vlan WIFI_CLIENT
```

Then configure the VLANs to assign the VLANs to a number and specific interfaces, then save the configuration:

```
configure vlan AP tag 10
```

```
configure vlan WIFI_CLIENT tag 20
```

```
configure vlan AP add ports 1 tagged
```

```
configure vlan AP add ports 2 untagged
```

```
configure vlan WIFI_CLIENT add ports 1 tagged
```

save configuration lab_user_name

Step 3: Configuring the Fortigate AP

1. Console into the FortiGate AP and configure it to the following settings.

```
FortiAP-421E # cfg -a ADDR_MODE=DHCP
```

```
FortiAP-421E # cfg -a AP_IPADDR=0.0.0.0
```

```
FortiAP-421E # cfg -a AP_NETMASK=0.0.0.0
```

```
FortiAP-421E # cfg -a IPGW=0.0.0.0
```

```
FortiAP-421E # cfg -a AP_MGMT_VLAN_ID=10
```

```
FortiAP-421E # cfg -a AC_DISCOVERY_TYPE=1
```

```
FortiAP-421E # cfg -a AC_IPADDR_1=(FG_LAN_IP)
```

```
FortiAP-421E # cfg -a AC_HOSTNAME_1=none
```

```
FortiAP-421E # cfg -a DNS_SERVER=(FG_LAN_IP)
```

```
FortiAP-421E # cfg -a WANLAN_MODE=WAN-ONLY
```

```
FortiAP-421E # cfg -c
```

```
FortiAP-421E # date -s "yyyy-mm-dd hh:mm:ss"
```

Step 4: Configuring the FortiGate 40f

1. Configure port A (interface connecting to switch) on the FortiGate 40f to have two sub interfaces/VLANs, corresponding to the VLANs on the switch.

a. VLAN 10 Setup

FortiGate-40F

Dashboard

Network

Interfaces

DNS

IPAM

FortiExtenders

SD-WAN

Static Routes

Diagnostics

Policy & Objects

Security Profiles

VPN

User & Authentication

WIFI & Switch Controller

System

Security Fabric

Log & Report

Edit Interface

Name: **AP Management**

Alias:

Type: **VLAN**

VLAN protocol: **802.1Q**

Interface: **a**

VLAN ID: **10**

Role: **LAN**

Address

Addressing mode: **Manual** IPAM DHCP PPPoE

IP/Netmask: **10.10.10.1/255.255.255.0**

Create address object matching subnet: ☒

Name: **AP Management address**

Destination: **10.10.10.0/24**

Secondary IP address: ☐

Administrative Access

IPv4

☐ HTTPS ☐ HTTP ☒ **PING**

☐ FMG-Access ☐ SSH ☐ SNMP

☐ FTM ☐ RADIUS Accounting ☒ **Security Fabric Connection**

☐ Speed Test ☐ SCIM

☒ **DHCP Server**

DHCP status: **Enabled** Disabled

Address range: **10.10.10.2-10.10.10.10**

Netmask: **255.255.255.0**

Default gateway: **Same as Interface IP** Specify

DNS server: **Same as System DNS** Same as Interface IP Specify

Lease time: **604800** second(s)

FortiClient On-Net Status: ☐

Advanced

Network

OK Cancel

FortiGate-40F

Status: **Up**

MAC address: **84:39:8f:f6:1f:b0**

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b. VLAN 20 Setup

New Interface

Name: WiFi Clients

Alias:

Type: VLAN

VLAN protocol: 802.1Q 802.1AD

Interface: a

VLAN ID: 20

Role: LAN

Addressing mode: Manual IPAM DHCP PPPoE One-Arm Sniffer

IP/Netmask: 10.10.20.1/24

Create address object matching subnet: ☒

Name: WiFi Clients address

Destination: 10.10.20.0/24

Secondary IP address: ☐

Administrative Access

IPv4: ☐ HTTPS ☐ HTTP ☒ PING ☐ SNMP ☐ Security Fabric Connection ☐ FMG-Access ☐ SSH ☐ RADIUS Accounting ☐ Speed Test ☐ SCIM

DHCP Server

DHCP status: ☒ Enabled ☐ Disabled

Address range: 10.10.20.2-10.10.20.254

Netmask: 255.255.255.0

Default gateway: Same as Interface IP Specify

DNS server: Same as System DNS Same as Interface IP Specify

Lease time: 604800 second(s)

Advanced

OK Cancel

FortiGate
FortiGate-40F

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Activate Windows
Go to Settings to activate Windows.

2. Create SSIDs for the AP. For this lab, two SSIDs are created, with one utilizing WPA2-Personal Pre-shared Passkey, and one utilizing WPA2-Enterprise Radius Server Authentication.

a. This creates the SSID for WPA2-Personal with Pre-shared Passkey:

FortiGate-40F

Dashboard > Network > Policy & Objects > Security Profiles > VPN > User & Authentication > **WiFi & Switch Controller** > Managed FortiAPs > WiFi Clients > WiFi Maps > **SSIDs** > FortiAP Profiles > WIDS Profiles > WiFi Settings > FortiLink Interface > Managed FortiSwitches > FortiSwitch Clients > FortiSwitch VLANs > FortiSwitch Ports > FortiSwitch Port Policies > NAC Policies > System > Security Fabric > Log & Report

Create New SSID

Name: **AYCyber PSK**

Alias:

Type: WIFI SSID

Traffic mode: **Tunnel** Bridge Mesh

Address

Addressing mode: **Manual** IPAM One-Arm Sniffer

IP/Netmask: **10.10.1.1/24**

Create address object matching subnet: ☒

Name: AYCyber PSK address

Destination: 10.10.1.0/24

Secondary IP address: ☐

Administrative Access

IPv4: ☐ HTTPS ☐ HTTP ☐ PING ☐ FMG-Access ☐ SSH ☐ SNMP ☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

☒ **DHCP Server**

DHCP status: **Enabled** Disabled

Address range: 10.10.1.2-10.10.1.254

Netmask: 255.255.255.0

Default gateway: **Same as Interface IP** Specify

DNS server: **Same as System DNS** Same as Interface IP Specify

Lease time: ☒ 604800 second(s)

Advanced

Network

Device detection: ☒

WiFi Settings

SSID: fortinet

OK Cancel

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SSID

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WiFi & Switch Controller

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WiFi Clients

WiFi Maps

SSIDs

FortiAP Profiles

WIDS Profiles

WiFi Settings

FortiLink Interface

Managed FortiSwitches

FortiSwitch Clients

FortiSwitch VLANs

FortiSwitch Ports

FortiSwitch Port Policies

NAC Policies

System

Security Fabric

Log & Report

Create New SSID

Network

Device detection

WiFi Settings

SSID

Client limit

Broadcast SSID

Beacon advertising

Name

Model

Serial number

Security Mode Settings

Security mode

Captive Portal

Pre-shared Key

Mode

Single

Multiple

Passphrase

Client MAC Address Filtering

RADIUS server

Address group policy

Disable

Allow

Deny

Additional Settings

Schedule

In Class Schedule

Optional

Block intra-SSID traffic

Optional VLAN ID

Broadcast suppression

ARPs for known clients

DHCP unicast

DHCP uplink

Quarantine host

VLAN pooling

NAC profile

Traffic Shaping

Outbound shaping profile

Outbound bandwidth

Miscellaneous

Comments

OK

Cancel

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3. Before you can create the SSID for WPA2-Enterprise, you first need to configure a radius server definition on the FG:

FortiGate-40F

Dashboard >

Network >

Policy & Objects >

Security Profiles >

VPN >

User & Authentication

User Definition

User Groups

Guest Management

LDAP Servers

RADIUS Servers

SCIM Clients

Single Sign-On

Authentication Settings

FortiTokens

WiFi & Switch Controller >

System >

Security Fabric 1 >

Log & Report >

FortiGate

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Edit RADIUS Server

Name: linuxRadiusServer

Authentication method: Default Specify

NAS IP:

Include in every user group: ☐

Primary Server

Radius Server IP Address

IP/Name: 192.168.10.4

Secret: Make sure to match the Secret on both ends

Connection status: Successful

Test Connectivity

Test User Credentials

Secondary Server

IP/Name:

Secret:

Test Connectivity

Test User Credentials

OK Cancel

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4. Now create the SSID for WPA2-Enterprise Radius Server Authentication:

Create New SSID

Name:

Alias:

Type: ☒ WiFi SSID

Traffic mode: ☒ Tunnel ☐ Bridge ☐ Mesh

Address

Addressing mode: ☒ Manual ☐ IPAM ☐ One-Arm Sniffer

IP/Netmask:

Create address object matching subnet: ☒

Name:

Destination:

Secondary IP address: ☐

Administrative Access

IPv4: ☐ HTTPS ☐ HTTP ☐ PING ☐ FMG-Access ☐ SSH ☐ SNMP ☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection ☐ Speed Test ☐ SCIM

☒ DHCP Server

DHCP status: ☒ Enabled ☐ Disabled

Address range:

Netmask:

Default gateway: ☒ Same as Interface IP ☐ Specify

DNS server: ☒ Same as System DNS ☐ Same as Interface IP ☐ Specify

Lease time: second(s)

☐ Advanced

Network

Device detection: ☒

WiFi Settings

SSID:

Client limit: ☐

Broadcast SSID: ☒

Beacon advertising: ☐ Name ☐ Model ☐ Serial number

Security Mode Settings

Security mode:

Authentication: ☒ RADIUS Server

Client MAC Address Filtering

RADIUS server: ☐

OK Cancel

5. Create a FortiAP Profile:

FortiGate-40F

Edit FortiAP Profile

Name: **FortiAP 421E**

Comments: Write a comment... 0/255

Platform: FAP421E

Indoor / Outdoor: **Default (Indoor)** Override

Country / Region: Use default (United States)

AP login password: Set **Leave Unchanged**

Administrative access: ☐ HTTPS ☐ SSH ☐ SNMP

Client load balancing: ☐ Frequency Handoff ☐ AP Handoff

802.1X authentication: ☐

Radio 1

Mode: Disabled **Access Point** Dedicated Monitor

WIDS profile: ☐

Radio resource provision: ☐

Band: 2.4 GHz 802.11n 802.11g

Channel width: 20MHz

Channel plan: **Three Channels** Four Channels Custom

Channels: ☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☒ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☒ 11

Short guard interval: ☐

Transmit power mode: ☒ Percent
Transmit power is determined by multiplying set percentage with maximum available power determined by region and FortiAP device.

☐ dBm
Power is setting using a dBm value.

☐ Auto
Set a range of dBm values and the power is set automatically.

Transmit power: 100 %

SSIDs: **Tunnel** Bridge **Manual**
AYCyber PSK (AYCyber PSK)

Monitor channel utilization: ☐

FortiGate

FortiGate-40F

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OK **Cancel**

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Add your premade SSID into the SSID Section

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Edit FortiAP Profile

Monitor channel utilization

Radio 2

Mode

Disabled

Access Point

Dedicated Monitor

WIDS profile

Radio resource provision

Band

5 GHz

802.11ac

802.11n

802.11a

Channel width

20MHz

40MHz

80MHz

Channels

Set Channels

36

40

44

48

52

56

60

64

100

104

108

112

116

120

124

128

132

136

140

144

149

153

157

161

Short guard interval

Transmit power mode

Percent

Transmit power is determined by multiplying set percentage with maximum available power determined by region and FortiAP device.

dBm

Power is setting using a dBm value.

Auto

Set a range of dBm values and the power is set automatically.

Transmit power

100 %

SSIDs

Tunnel

Bridge

Manual

AYCyber Radius (AYCyber Radius)

Monitor channel utilization

LAN Port

Port mode

Uplink Only

Uplink & Bridge

Mesh Setup

Broadcast mesh provisioning SSID

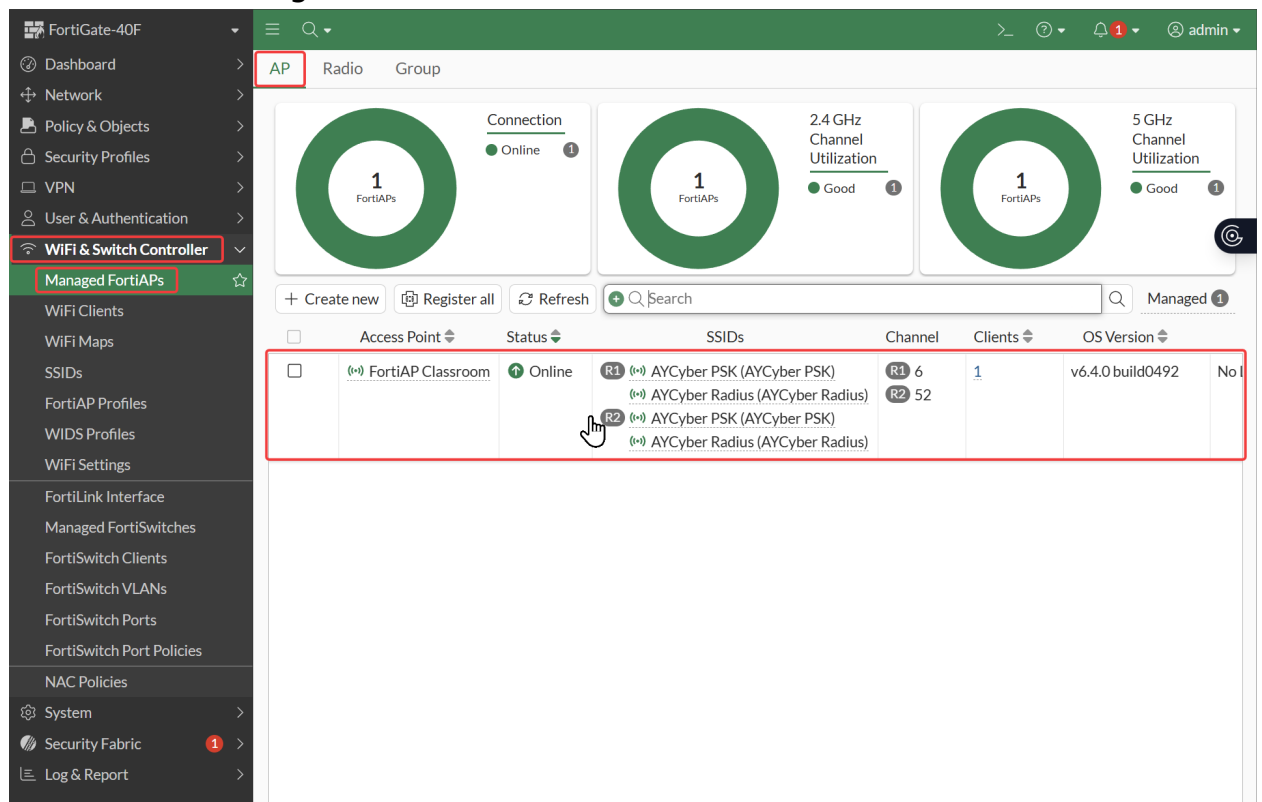
System Log

Syslog profile

OK

Cancel

6. Now you should be able to see your FortiAP under WiFi & Switch Controller -> Managed FortiAPs:



If it does not show, you could try to set it manually by after pressing “Create New” and using the AP’s serial number, which can be found when you run “cw_diag -c wtp-cfg” inside the AP’s CLI.

After the Managed FortiAPs screen show your AP as Online, you should be able to see the SSIDs being broadcast as WiFi on devices in range of AP now.

Problems:

There were multiple issues that was encountered. One of which was enabling the CAPWAP protocol on ports connecting to the AP. Initially when I only set the AP Management Sub-interface to Security Fabric Protocol (which includes CAPWAP Protocol) and the AP was being detected and discovered by the firewall, but it was not recognizing the AP as a managed FortiAP. The solution we first tried was adding a security policy to allow the AP Management Sub-Interface to specifically allow traffic to other interfaces, but that didn’t solve the issue, and then we realized that we need to enabled CAPWAP on all related interfaces, which included the A interface, and that was what resolved our issue.

Conclusions:

In this lab, we successfully configured an AP remotely through a FortiGate 40f Firewall. The configuration process began with the initial configuration and setup of the RADIUS server on a LINUX server and the VLAN separation on the Switch, and then we moved on to the FortiGate configuration. Inside the FortiGate, we configured multiple SSIDs for the AP, each with different security protocols also applied an optional schedule to automatically turn the AP SSID broadcasting on or off, as well as a FortiAP Profile which is what actually the settings that was being applied to the FortiAP is based on. After resolving a severe issue that was hindering progress, we were finally able to get the firewall to recognize the fortiaP as a managed FortiAP and then tested the internet connection by connecting to it through both lan devices and mobile devices not directly connected to the network.